

The question

How much of our headline contact-prediction number survives if we throw out every eval protein that has a sequence homolog in the training set?

The worry is concrete. Our training labels are contacts read off **AlphaFold2 (AFDB) and ESMFold2 (ESM Atlas) structures** — predictors that had MSAs or a large PLM. MarFold sees only a single sequence at inference, but if an eval protein has a close relative in the training corpus, the coevolution signal that produced that relative's structure is baked into the weights and reachable by homology. In that regime "single-sequence contact prediction" is partly an MSA-derived retrieval.

What was new here

Two prior experiments covered half of this. #65 ran an MMseqs2 leakage check, but only against the 1.33M AFDB cluster *representatives*, and never tied it to an accuracy number. #94 built the identity-stratified accuracy curve, but against the AFDB corpus alone and for the #61 model — 0.26 R-precision weaker than today's.

Neither had ever looked at the **66.8M ESM-Atlas sequences**, which are 94% of exp199's training set by count and are metagenomic — exactly the sequences most likely to supply remote homologs for natural eval proteins.

So: decode all 70.9M training sequences out of the two published corpora (neither carries a `sequence` column), search all 554 eval proteins against them, and re-aggregate the per-protein scores that already exist for six predictors. No new model inference.

Result 1 — the eval set is homology-rich

58% of the 554 eval proteins have a significant training homolog; only 11% have no MMseqs2 alignment at all. ESM-Atlas hits ****296**** proteins, more than AFDB's 275, and 48 proteins have a training relative that exists **only** in the metagenomic corpus. The half nobody had checked does matter.

But sequence novelty is not fold novelty: of the 231 sequence-novel proteins, 133 are still `same\_fold` and 61 `redundant` against the AFDB training folds. Only ****37**** are novel in both sequence and fold.

Result 2 — accuracy does not track training proximity

Spearman rho between best training identity and R-precision:

- MarFold #199: ****−0.12**** pooled, ****+0.04**** on natural proteins - seq-KNN k=10 (the copy-the-neighbour null): ****+0.53****

seq-KNN is the calibration. A model whose score came from retrieving memorised homologs would look like that last number. MarFold does not. Restricting to the 231 homology-free proteins moves R-precision 0.611 → 0.549, and against ESMFold/ESMFold2 on natural proteins MarFold actually **gains** ground.

This is #94's finding, extended to the full training set and a far stronger model.

Result 3 — but the Protenix single-seq parity does not survive

#180 records Marifold at parity with Protenix-v2 single-sequence: $+0.008$ [-0.017 , $+0.033$] over 554 proteins. That parity is homology-dependent.

- On proteins **with** a training homolog, Marifold leads by **$+0.13$** - On proteins **without** one, it trails by **-0.17** - Difference of differences: **-0.303** [-0.346 , -0.259], $\sim 100\times$ the tracker's 0.0023 noise floor

On the 37 proteins novel in both sequence and fold, Marifold scores **0.328** against Protenix-SS's 0.547 and ESMFold2's 0.620 .

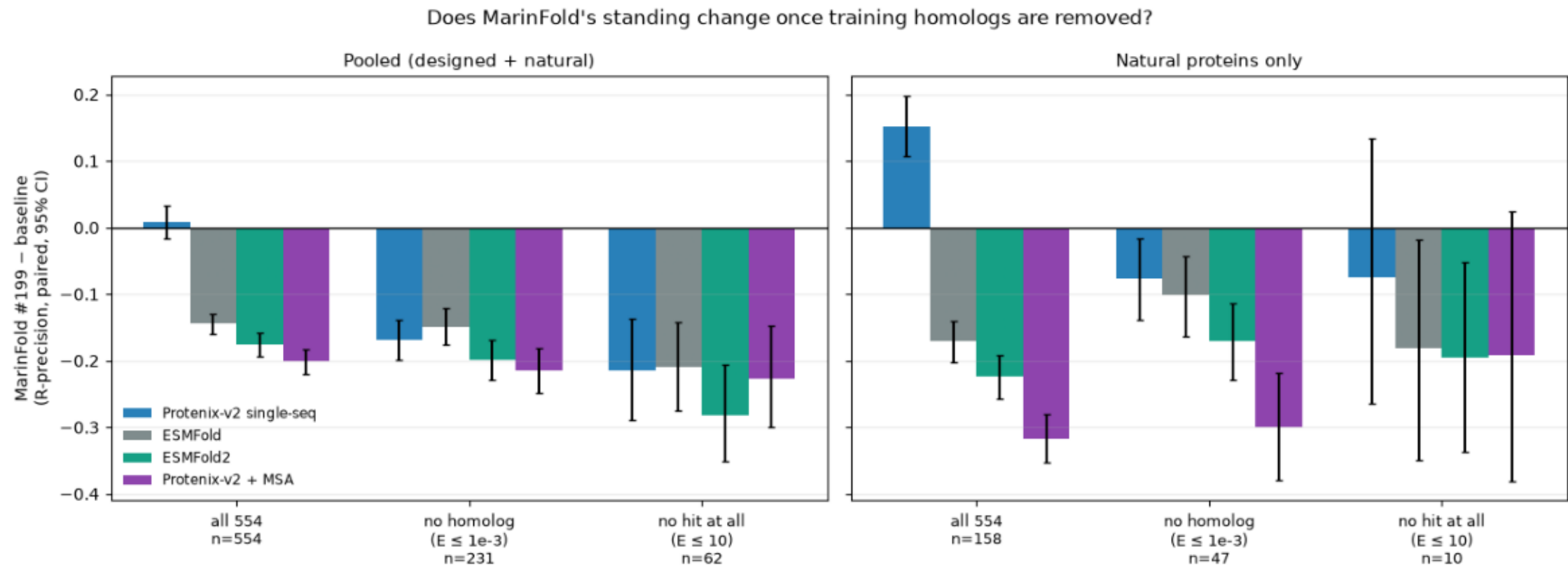
The effect is specific to Protenix-SS — Marifold's standing against ESMFold, ESMFold2 and Protenix+MSA does not degrade. That asymmetry is the opposite of what uniform homology leakage predicts, and points at Protenix-SS being differentially strong in the novel-fold regime rather than at Marifold leaking.

What to do with it

1. **The headline number is not substantially inflated by homology leakage** — but **"we have caught up with single-sequence Protenix"** should be qualified as "on proteins resembling our training data" until the novel-fold gap closes. 2. **Report the homology-free subset alongside the headline.** `data/eval_train_identity.csv` is committed per protein and joins on `(dataset, stem)` — one merge. 3. **This eval set cannot answer the question much better than it just did.** The homology-free subsets are 80% de novo designs; the natural, homology-free, fold-novel corner is $n \approx 19$. Measuring novel-protein performance rather than bounding it needs more natural low-homology proteins — the gap #65 was originally opened to fill.

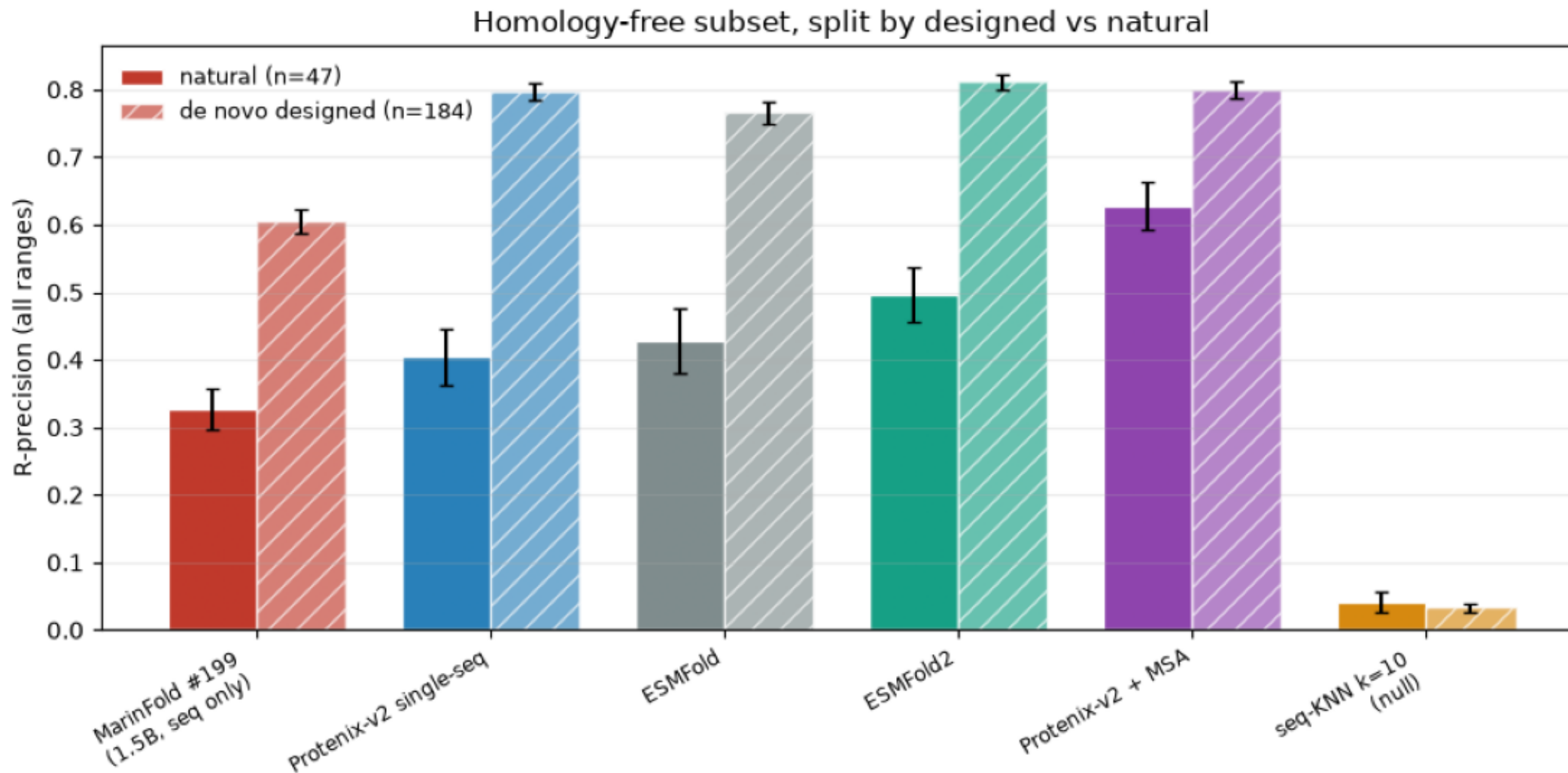
delta_across_subsets.png

Paired MarInFold-minus-baseline R-precision as the eval set is restricted to proteins with no training homolog. Above zero means MarInFold wins. Bars are 95% percentile CIs from 10,000 paired bootstrap resamples over proteins. The right panel drops the de novo designs, which make up 80-92% of the homology-free subsets and which structure predictors find unusually easy.



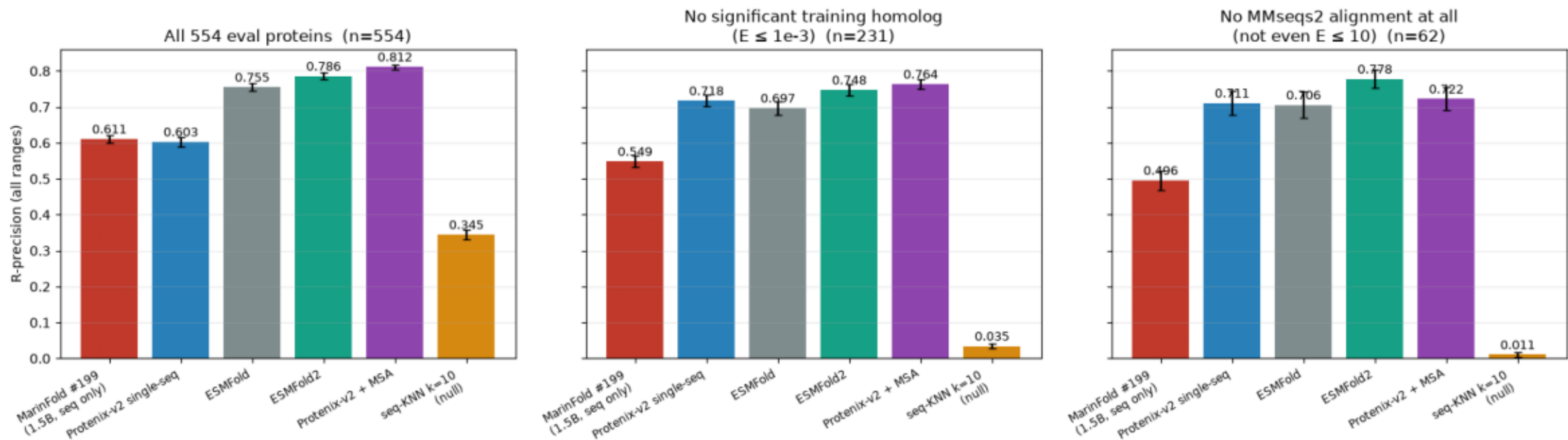
designed_vs_natural.png

The homology-free subset is not homogeneous: de novo designed proteins have no homologs anywhere by construction, and structure predictors find their idealised backbones easy. Splitting the subset keeps that confound visible.



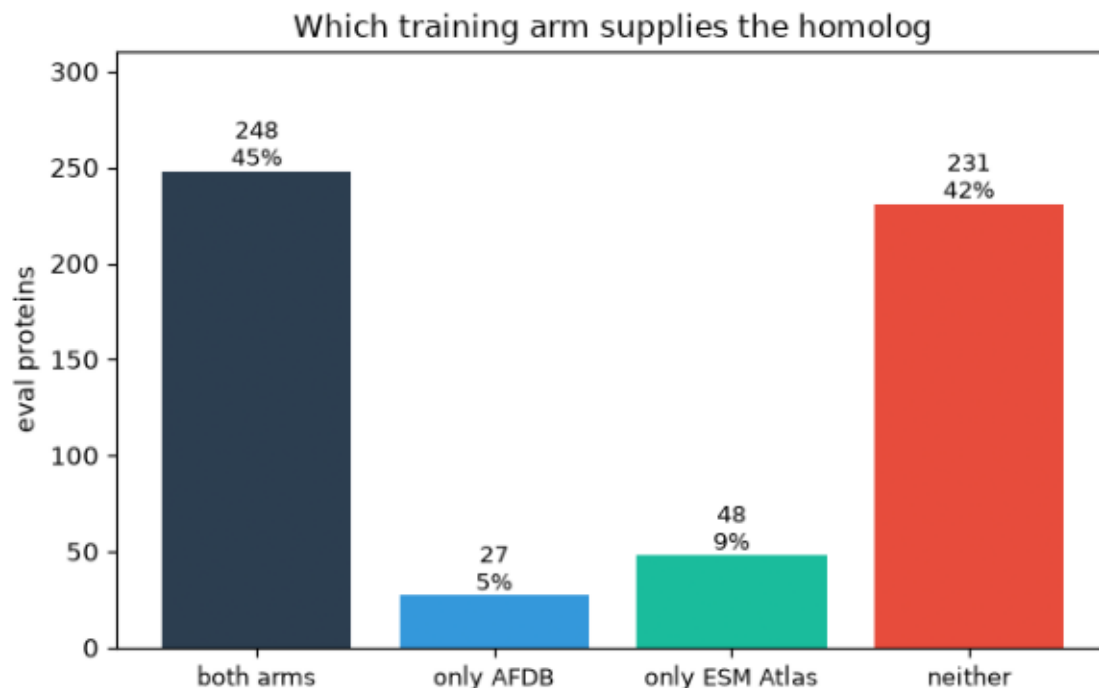
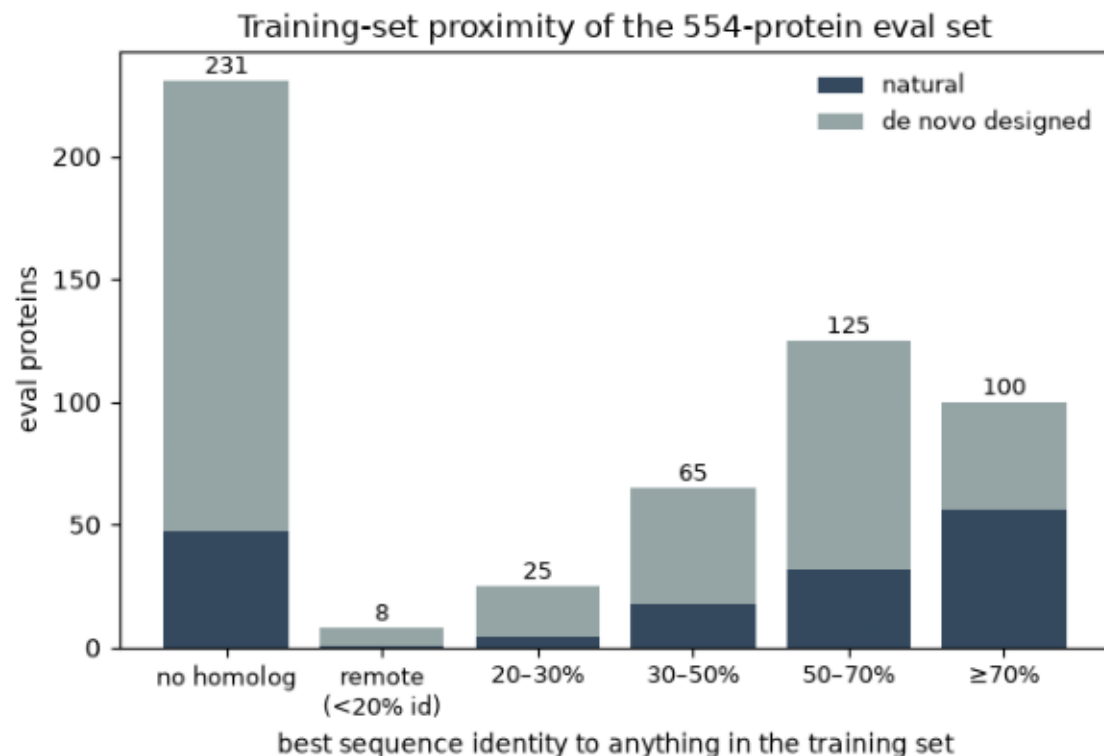
headline_homology_free.png

The pre-registered cut. Left: the published benchmark. Middle and right: only eval proteins with no homolog in either training corpus, under the conventional significance line and under the strictest possible one — shown together because that threshold is the experiment's one judgement call. Error bars are $\pm$ SEM over proteins; paired MarInFold-minus-baseline CIs are in data/headline.csv.



overlap_profile.png

Left: eval proteins by best sequence identity to exp199's training set (MMseqs2 -s 7.5; identity measured over alignments covering $\geq 50\%$ of the query), split on de novo designed vs natural. Right: which of the two training corpora contributes the homolog. 'neither' is the homology-free subset the re-eval uses.



rprecision_vs_identity.png

R-precision (all ranges, $\pm$ SEM) across training-identity strata, all six predictors on the same proteins. seq-KNN is the positive control — a pure copy-the-neighbour model must track identity. A flat MarinFold curve means its skill is not homology retrieval.

